

Comparing Unlike Fractions

Answer Key

Grade 4–5

Quick Answers

- | | |
|--|---|
| 1. 4 | $7/9 = ?/18 = 14$, comparison symbol = $>$ |
| 2. $<$ | 7. comparison symbol = $<$, — |
| 3. numerator for $2/3 = ?/12 = 8$, numerator for $3/4 = ?/12 = 9$, comparison symbol = $<$ | 8. (a) numerator for $2/3 = ?/12 = 8$, (b) comparison symbol = $=$ |
| 4. $\frac{4}{10}$ | 9. numerator for $7/12 = ?/24 = 14$, numerator for $5/8 = ?/24 = 15$, comparison symbol = $<$ |
| 5. Jug A against one half = $>$, Jug B against one half = $<$ | 10. (a) comparison symbol = $<$, — |
| 6. numerator for $5/6 = ?/18 = 15$, numerator for | |
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What is verified

8 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4–5

4.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 20 tagged checks.

1. The strip is cut into 8 equal parts and half is shaded. Fill in: $\frac{1}{2} = \frac{\square}{8}$

Solution: Half of the 8 parts are shaded, and half of 8 is $8 \div 2 = 4$. So $\frac{1}{2} = \frac{4}{8}$; the missing numerator is 4

2. Write the correct symbol: $\frac{3}{8} \text{ -- } \frac{1}{2}$

Solution: The benchmark $\frac{1}{2}$ written in eighths is $\frac{4}{8}$. Since 3 eighths is less than 4 eighths, we get $\frac{3}{8} \text{ } \boxed{<} \text{ } \frac{1}{2}$

3. Compare $\frac{2}{3}$ and $\frac{3}{4}$ by rewriting both in twelfths.

Solution: $\frac{2}{3}$: multiply top and bottom by 4 to get twelfths — numerator 8

$\frac{3}{4}$: multiply top and bottom by 3 — numerator 9

Now compare 8 twelfths with 9 twelfths: 8 is smaller, so $\frac{2}{3} \text{ } \boxed{<} \text{ } \frac{3}{4}$

4. Lena's ribbon is $\frac{2}{5}$ m; her ruler is marked in tenths. Write the same length in tenths of a meter.

Solution: Multiply top and bottom by 2: $\frac{2}{5} = \frac{2 \times 2}{5 \times 2}$, which is $\boxed{\frac{4}{10}}$

5. Jug A is $\frac{5}{8}$ full, Jug B is $\frac{3}{8}$ full. Compare each to $\frac{1}{2}$.

Solution: One half in eighths is $\frac{4}{8}$. Jug A: 5 eighths is more than 4 eighths, so $\frac{5}{8} \boxed{>} \frac{1}{2}$
Jug B: 3 eighths is less than 4 eighths, so $\frac{3}{8} \boxed{<} \frac{1}{2}$ Only Jug A is more than half full.

6. Pedro ran $\frac{5}{6}$ mi, Amara ran $\frac{7}{9}$ mi. Rewrite both in eighteenths and compare.

Solution: $\frac{5}{6}$: multiply top and bottom by 3 — numerator

$\boxed{15}$

$\frac{7}{9}$: multiply top and bottom by 2 — numerator

$\boxed{14}$

15 eighteenths beats 14 eighteenths, so $\frac{5}{6} \boxed{>} \frac{7}{9}$ — Pedro ran farther.

7. Compare $\frac{4}{9}$ and $\frac{6}{11}$ using the benchmark $\frac{1}{2}$, and explain.

Solution: Half of 9 is 4.5, and $4 < 4.5$, so $\frac{4}{9}$ is *less* than $\frac{1}{2}$. Half of 11 is 5.5, and $6 > 5.5$, so $\frac{6}{11}$ is *more* than $\frac{1}{2}$. A fraction below the benchmark is smaller than one above it: $\frac{4}{9} \boxed{<} \frac{6}{11}$

Explanation (instructor judges): full credit for any reasoning that places $\frac{4}{9}$ below one half and $\frac{6}{11}$ above one half and concludes from the benchmark.

8. Marco shades $\frac{2}{3}$ of one strip and $\frac{8}{12}$ of an identical strip. (a) Fill in $\frac{2}{3} = \frac{\square}{12}$. (b) Compare.

Solution: (a) Multiply top and bottom by 4: the numerator is

$\boxed{8}$

(b) Both strips show 8 twelfths, so the fractions are equivalent: $\frac{2}{3} \boxed{=} \frac{8}{12}$

9. Jon has $\frac{7}{12}$ of a pan, Kayla has $\frac{5}{8}$. Rewrite both in twenty-fourths and compare.

Solution: $\frac{7}{12}$: multiply top and bottom by 2 — numerator

$\boxed{14}$

$\frac{5}{8}$: multiply top and bottom by 3 — numerator

$\boxed{15}$

14 twenty-fourths is less than 15 twenty-fourths, so $\frac{7}{12} \boxed{<} \frac{5}{8}$ — Kayla has more.

10. Challenge. (a) Compare $\frac{3}{8}$ and $\frac{1}{2}$. (b) Find a fraction between them and explain.

Solution: (a) One half is $\frac{4}{8}$, and 3 eighths is less: $\frac{3}{8} \boxed{<} \frac{1}{2}$

(b) *(instructor judges)* Any fraction strictly between works. Example: in sixteenths, $\frac{3}{8} = \frac{6}{16}$ and $\frac{1}{2} = \frac{8}{16}$, so $\frac{7}{16}$ fits between them. Another correct choice is $\frac{2}{5}$. Credit any valid fraction with sound reasoning.